

Introduction to Machine Learning

What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed.

Traditional programming follows:

Rules + Data → Output

Machine learning follows:

Data + Output examples → Learned Model

Types of Machine Learning

1. Supervised Learning

Supervised learning uses labeled datasets where each training example contains input data and the correct output.

Examples:

- Email spam detection
- House price prediction
- Image classification

Common algorithms:

- Linear Regression
 - Logistic Regression
 - Decision Trees
 - Support Vector Machines
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2. Unsupervised Learning

Unsupervised learning works with data that does not have predefined labels. The goal is to discover hidden patterns.

Examples:

- Customer segmentation
- Anomaly detection
- Data compression

Common algorithms:

- K-Means Clustering

- Principal Component Analysis (PCA)
 - Hierarchical Clustering
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3. Reinforcement Learning

Reinforcement learning trains an agent to make decisions by interacting with an environment.

The agent receives:

- Rewards for good actions
- Penalties for bad actions

Examples:

- Game-playing AI
 - Robotics
 - Autonomous vehicles
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Neural Networks

A neural network is a computing model inspired by the human brain.

A neural network contains:

Input Layer

Receives the raw data.

Example:

Image pixels → Input Layer

Hidden Layers

Perform mathematical transformations and learn features.

More hidden layers create deep neural networks.

Output Layer

Produces the final prediction.

Deep Learning

Deep Learning is a subset of machine learning that uses neural networks with many layers.

Applications include:

- Speech recognition
- Computer vision

- Natural language processing
- Generative AI

Popular deep learning frameworks:

- TensorFlow
- PyTorch

Natural Language Processing

Natural Language Processing (NLP) allows computers to understand and generate human language.

Common NLP tasks:

Text Classification

Assigning categories to text.

Example:

"I love this product"

→ Positive sentiment

Text Generation

Creating new text using language models.

Examples:

- Chatbots
- AI assistants
- Translation systems

Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation combines information retrieval with language generation.

A basic RAG pipeline:

User Query

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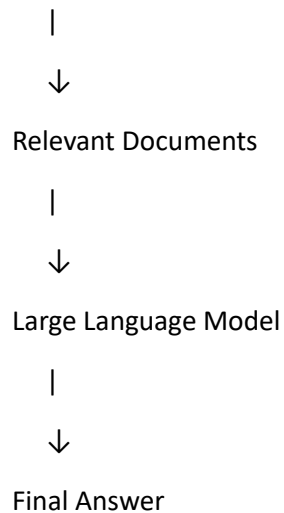
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Embedding Generation

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Vector Database Search



Benefits:

- Uses external knowledge
- Reduces hallucination
- Allows private document search

Vector Embeddings

Embeddings convert text into numerical vectors.

Example:

"cat"



[0.21, 0.84, -0.12, ...]

Similar meanings produce similar vectors.

Vector databases store and search these embeddings efficiently.

Examples:

- FAISS
- ChromaDB
- Pinecone

Chunking in RAG

Large documents are divided into smaller pieces called chunks.

Example:

Large PDF

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Chunk 1

Chunk 2

Chunk 3

Good chunking improves retrieval accuracy.

Common strategies:

- Fixed-size chunks
- Sentence-based chunks
- Paragraph-based chunks

Summary

Machine learning enables computers to learn from data. Deep learning uses neural networks for complex tasks. RAG combines retrieval systems with language models to provide accurate answers from external documents.